



PHILADELPHIA
— SEPTEMBER 27-30, 2026 —
Rooted in Brotherly Love & Lifelong Community

ANNUAL
MEETING

From NICU to Clinic: Translating Neuroprotection Research

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**Research
Saves Lives**





Speaker Name: Lauren Leigh Cooney Jantzie, PD

1. Disclosure of Relevant Financial Relationships

I have no financial relationships to disclose.

2. Disclosure of Off-Label and/or investigative uses:

I will discuss the following off label use and/or investigational use in my presentation:

- IND approved, off label use of erythropoietin (EPO) and melatonin (MLT)
- *Safety of combined therapy with erythropoietin and melatonin for preterm infants with intraventricular hemorrhage (SCEMPI)*
 - NCT05617833
 - IRB00301237
 - 1R01HD104673-01A1
 - PCT/US2022/018046

**Research
Saves Children**



Neurorepair & Neurorestoration for Cerebral Palsy

- Many children are diagnosed with cerebral palsy after the newborn period when they are no longer eligible for perinatal interventions

Perinatal Period

Term



magnesium
steroids

Diagnosis of CP

2 years

**Intervention to
Restore
Neurodevelopment
at Typical Age of
Diagnosis**

Childhood Interventions

- Physical Therapy
- Botulinum injections
- Orthotics
- Selective Dorsal Rhizotomy
- Intrathecal Baclofen Therapy
- Orthopedic Surgery

*** All much less effective in setting of inflammation**

Neurorestoration of Microenvironment to Improve Function, Comfort and Independence

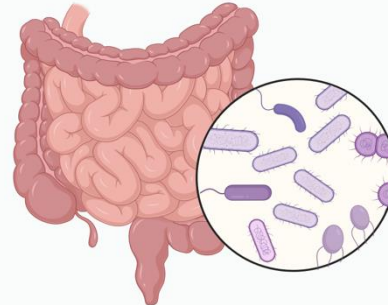
Treatment Options After the Neonatal Period



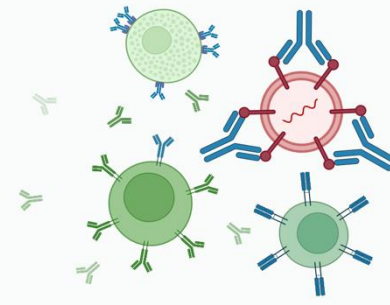
Caffeine



Melatonin



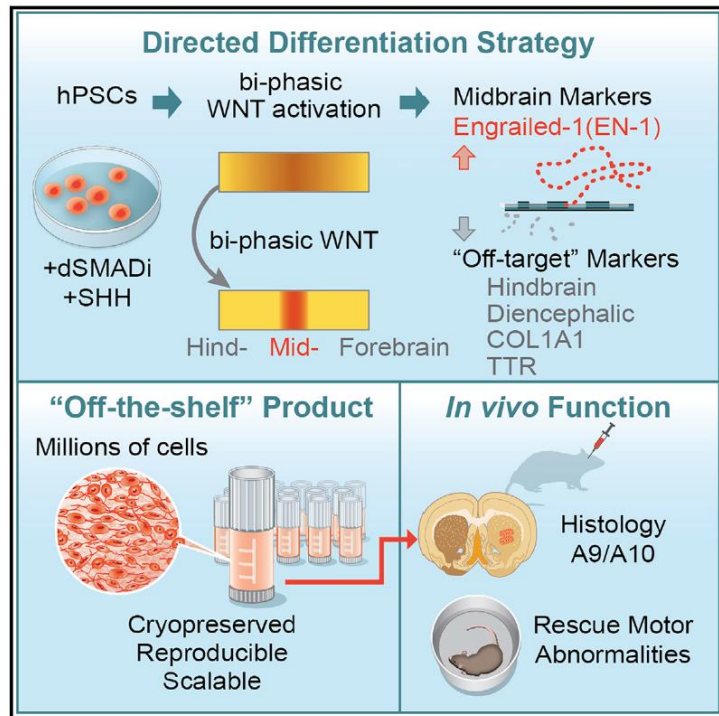
Microbiome Optimization/
Probiotics



Novel Immunomodulation

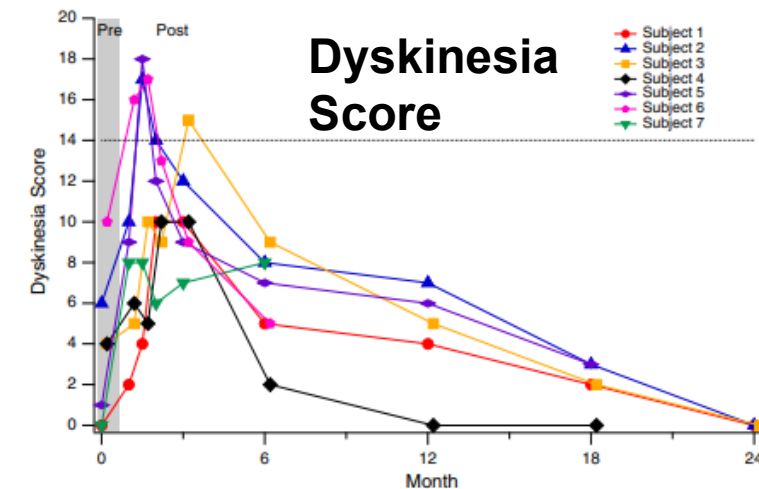
Additional Solutions

- Cell-based & Gene Therapies
 - Specific enzyme deficiencies
 - Parkinson's Disease, Epilepsy



Gene therapy for aromatic L-amino acid decarboxylase deficiency by MR-guided direct delivery of AAV2-AADC to midbrain dopaminergic neurons

Toni S. Pearson^{1,2,12}, Nalin Gupta^{1,12}, Waldy San Sebastian¹, Jill Imamura-Ching¹, Amy Viehoveer^{1,3}, Ana Grijalvo-Perez³, Alex J. Fay³, Neha Seth⁴, Shannon M. Lundy⁵, Youngho Seo^{1,6}, Miguel Pampaloni⁶, Keith Hyland⁷, Erin Smith⁸, Gardenia de Oliveira Barbosa⁹, Jill C. Heathcock⁹, Amy Minnema¹⁰, Russell Lonser¹⁰, J. Bradley Elder¹⁰, Jeffrey Leonard^{10,11}, Paul Larson¹ & Krystof S. Bankiewicz^{1,10}



Kim, Cell Stem Cell 2021
Pearson, Nat Comm 2021
Mcdowell, Res Dev Dis 2017

Recent Progress

- Early diagnosis - Determining who is at risk

Network Implementation of Guideline for Early Detection Decreases Age at Cerebral Palsy Diagnosis

Nathalie L. Maitre, MD, PhD,^{a,h} Vera J. Burton, MD, PhD,^{c,d} Andrea F. Duncan, MD, MSClinRes,^e Sai Iyer, MD,^f Betsy Ostrander, MD,^g Sarah Winter, MD,^g Lauren Ayala, DPT,^g Stephanie Burkhardt, MPH,^a Gwendolyn Gerner, PsyD,^{c,d} Ruth Getachew, BS,^c Kelsey Jiang, BS,^f Laurie Leshner, RN, MBA,^g Carrie M. Perez, MA, LPA,^e Melissa Moore-Clingenpeel, MA, MAS,^b Rebecca Lam, BA,ⁱ Dennis J. Lewandowski, PhD,^a Rachel Byrne, PTⁱ

JAMA Pediatrics | Review

Early Intervention for Children Aged 0 to 2 Years
With or at High Risk of Cerebral Palsy
International Clinical Practice Guideline Based on Systematic Reviews

Catherine Morgan, PhD; Linda Fethers, PhD; Lars Adde, PhD; Nadia Badawi, PhD; Ada Bancalé, NPT;
Roslyn N. Boyd, PhD; Olena Chorna, CCRP; Giovanni Cioni, MD; Diane L. Damiano, PhD; Johanna Damrah, PhD;
Linda S. de Vries, PhD; Stacey Dusing, PhD; Christa Einspieler, PhD; Ann-Christin Eliasson, PhD;
Donna Ferriero, MD; Darcy Fehlings, MD; Hans Forsberg, MD; Andrew M. Gordon, PhD; Susan Greaves, PhD;
Andrea Guzzetta, PhD; Mijna Hadders-Algra, PhD; Regina Harbourne, PhD; Petra Karlsson, PhD;
Lena Krumlinde-Sundholm, PhD; Beatrice Latal, PhD; Alison Loughran-Fowlds, PhD; Catherine Mak, PhD;
Nathalie Maitre, MD; Sarah McIntyre, PhD; Cristina Mei, PhD; Angela Morgan, PhD;
Angelina Kakooza-Mwesige, PhD; Domenico M. Romeo, PhD; Katherine Sanchez, PhD; Alicia Spittle, PhD;
Roberta Shepherd, PhD; Marelle Thornton, DipEd; Jane Valentine, PhD; Roslyn Ward, PhD;
Koa Whittingham, PhD; Alieh Zamany, DPT; Iona Novak, PhD

Maitre et al, Pediatrics 2020
Morgan et al, JAMA 2021

Genetic Testing

The importance of de novo mutations for pediatric neurological disease—It is not all in utero or birth trauma☆

Robert P. Erickson*

Clinically relevant copy number variations detected in cerebral palsy

Maryam Oskoui^{1,*}, Matthew J. Gazzellone^{2,3,*}, Bhooma Thiruvahindrapuram^{2,3}, Mehdi Zarrei^{2,3}, John Andersen^{4,5}, John Wei^{2,3}, Zhuozhi Wang^{2,3}, Richard F. Wintle^{2,3}, Christian R. Marshall^{2,3,6}, Ronald D. Cohn^{3,7,8,9}, Rosanna Weksberg^{3,7,9,10}, Dimitri J. Stavropoulos^{6,11}, Darcy Fehlings¹², Michael I. Shevell¹ & Stephen W. Scherer^{2,3,8,13}

IMMEDIATE COMMUNICATION

Whole-exome sequencing points to considerable genetic heterogeneity of cerebral palsy

G McMichael¹, MN Bainbridge², E Haan^{3,4}, M Corbett^{1,4}, A Gardner^{1,4}, S Thompson^{4,5}, BWM van Bon^{3,6}, CL van Eyk¹, J Broadbent¹, C Reynolds¹, ME O'Callaghan¹, LS Nguyen⁴, DL Adelson⁷, R Russo⁸, S Jhangiani², H Doddapaneni², DM Muzny², RA Gibbs², J Gecz^{1,4,9,10} and AH MacLennan^{1,9,10}

Cerebral palsy: causes, pathways, and the role of genetic variants

Alastair H. MacLennan, MD, FRANZCOG; Suzanna C. Thompson, MBBS, FRACP; Jozef Gecz, PhD

Erickson, Mut Res 2016
Oskoui et al, Nat Comm 2015
McMichael et al, Mol Psych 2015
MacLennan et al, Am J OBGyn 2015

It's time to offer genetic testing to all individuals diagnosed with cerebral palsy

Andrés Moreno-De-Luca

Autism & Developmental Medicine Institute, Department of Radiology,

JAMA Pediatrics | Original Investigation

Diagnostic Yield of Exome Sequencing in Cerebral Palsy and Implications for Genetic Testing Guidelines A Systematic Review and Meta-analysis

Pedro J. Gonzalez-Mantilla, MD; Yirui Hu, PhD; Scott M. Myers, MD; Brenda M. Finucane, MS;
David H. Ledbetter, PhD; Christa L. Martin, PhD; Andres Moreno-De-Luca, MD, MBA

Predictors of whole exome sequencing in dystonic cerebral palsy and cerebral palsy-like disorders

P. Pavelekova^{a,*}, J. Necpal^{b,c}, R. Jech^d, P. Havrankova^d, J. Svantnerova^c, V. Jurkova^e,
Z. Gdovinova^{a,f}, A. Lackova^{a,f}, V. Han^{a,f}, J. Winkelmann^{g,h,i,j}, M. Zech^{g,h}, M. Skorvanek^{a,f}

All Patients With a Cerebral Palsy Diagnosis Merit Genomic Sequencing

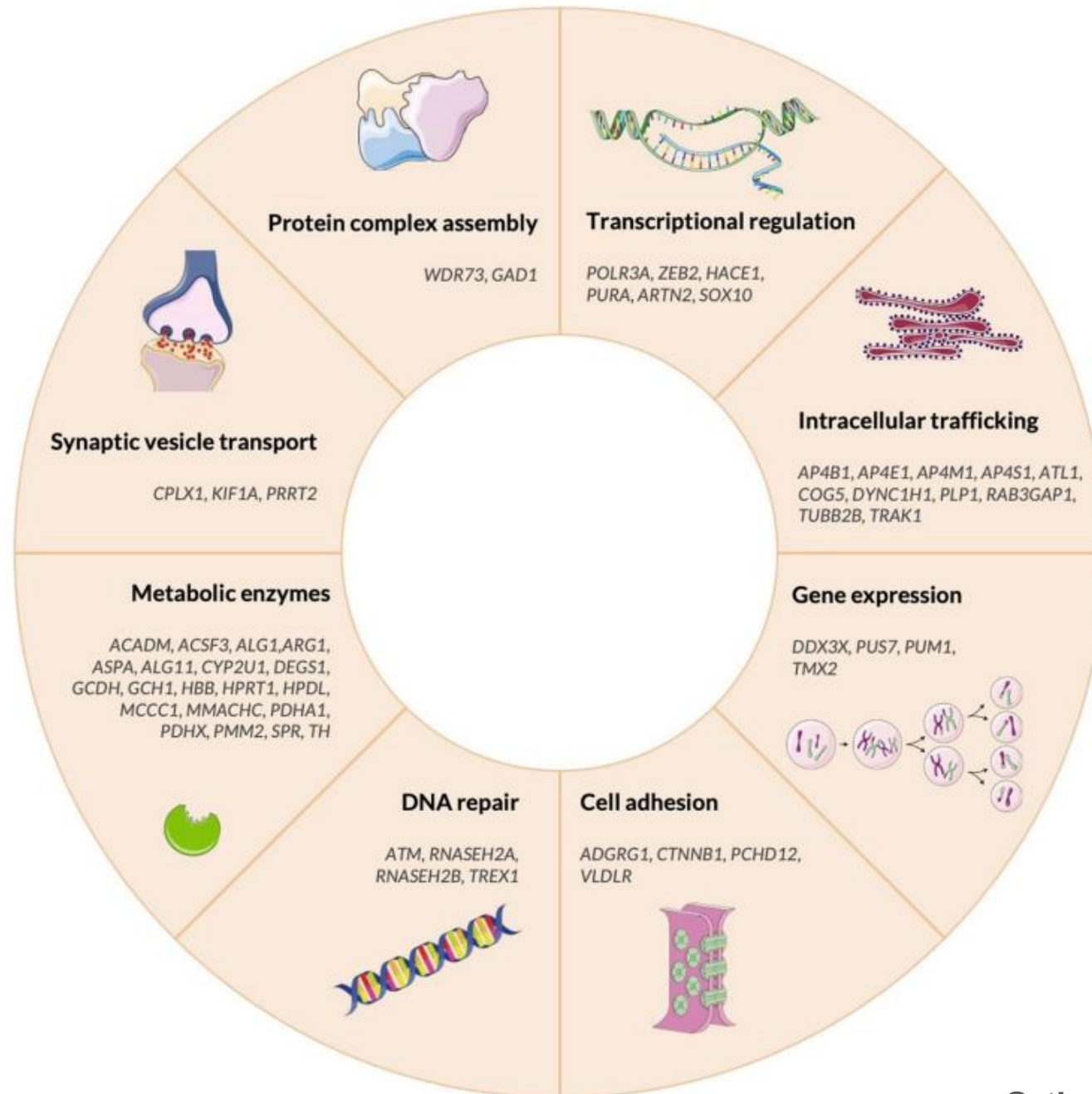
Clare van Eyk, PhD; Suzanna C. MacLennan, MB, BS; Alastair H. MacLennan, AO, MB, ChB, MD

Redefining cerebral palsies as a diverse group of neurodevelopmental disorders with genetic aetiology

Clare L. van Eyk^{1,2}, Michael C. Fahey³ & Jozef Gecz^{1,2,4}

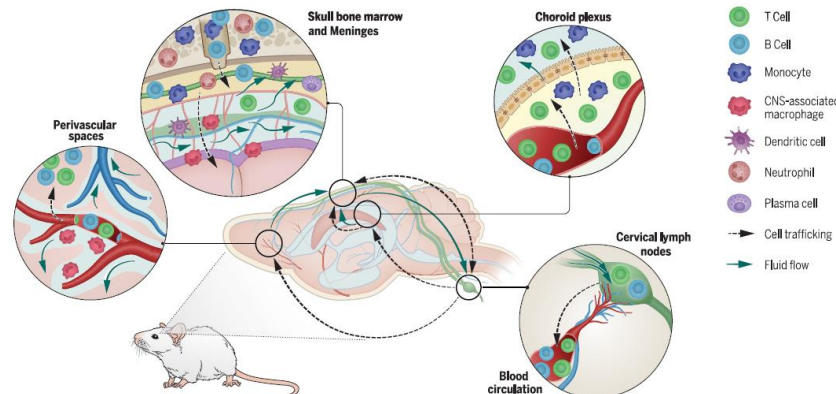
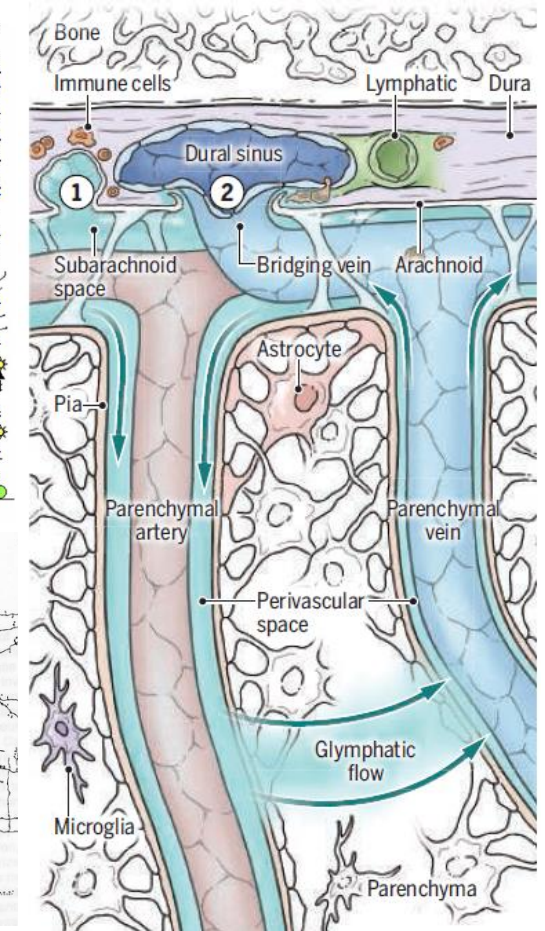
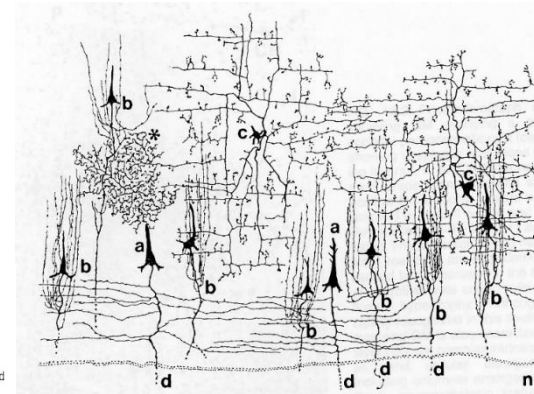
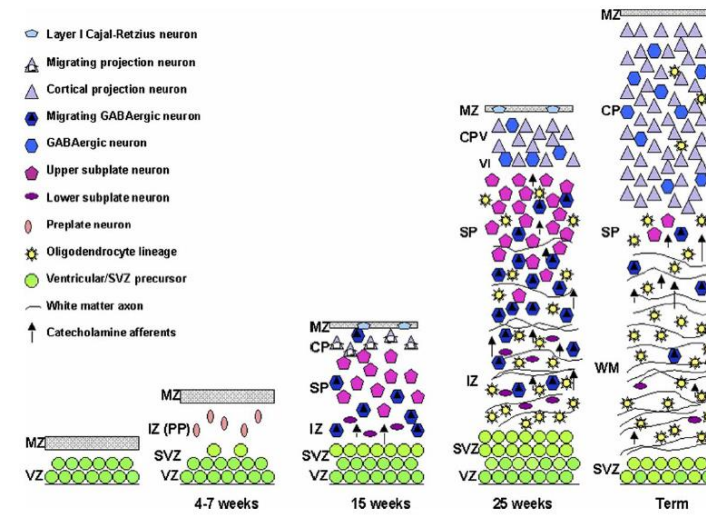
Moreno-De-Luca, PD & Rel Dis 2023
Gonzalez-Mantilla et al, JAMA Peds 2023
Pavelekova et al, PD & Rel Dis 2023
van Eyck et al, Nat Rev Neuro 2023

Mutations in CP occur across wide array of cellular functions



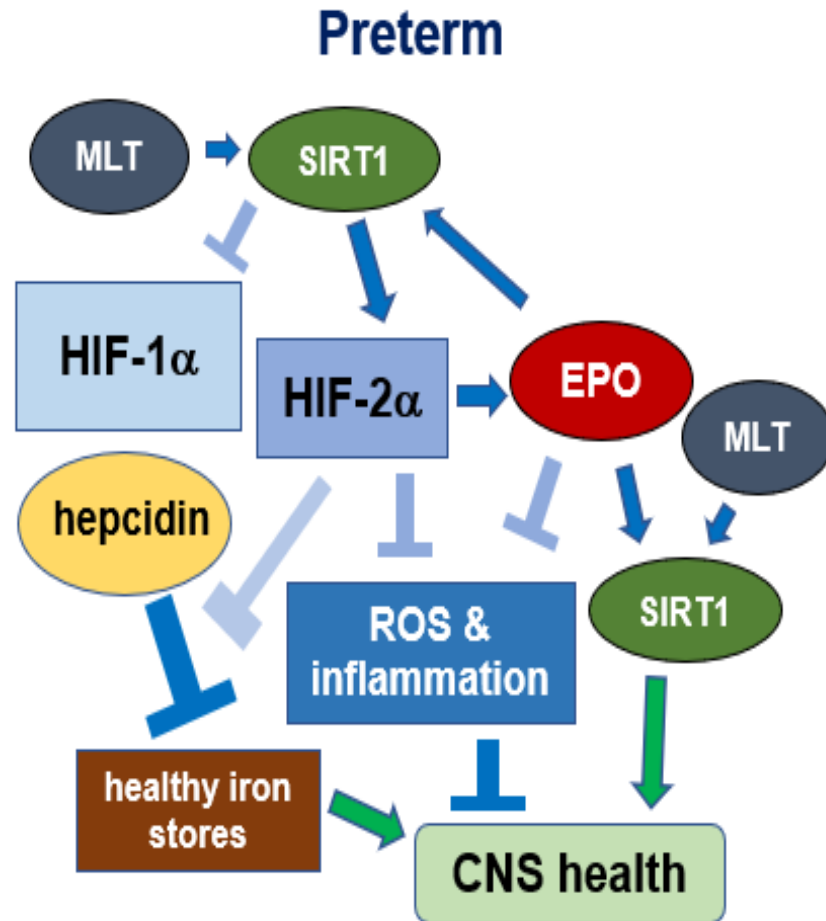
Damage to the Developing Brain

- White matter loss
 - Failure of new waves to replenish lost neural cells
- Axonal loss
 - Loss of most fiber tracts
- Gray matter loss
 - Disturbed synaptogenesis
 - Impaired myelination
- Defective waste clearance
- Impaired immune cell function



Marin-Padilla, Adv Neuro 2000
 Robinson Epilepsy & Beh 2005
 Castellani Science 2023
 Kipnis Science 2024

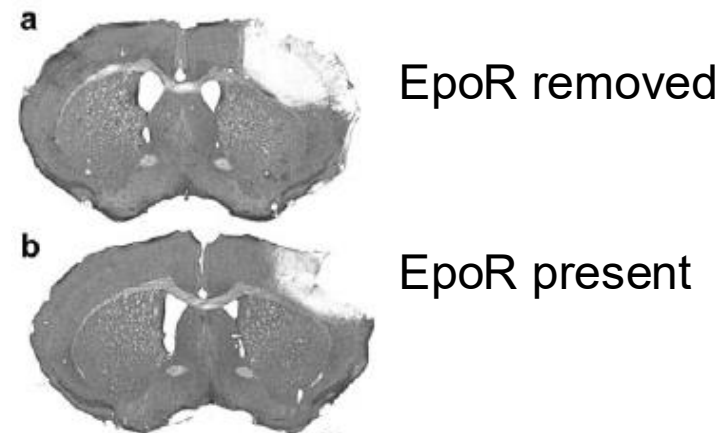
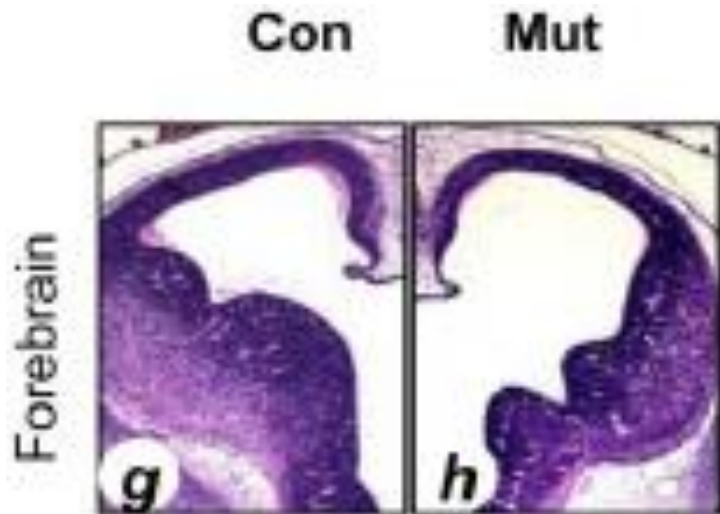
Hypothesis of MLT and EPO Synergy



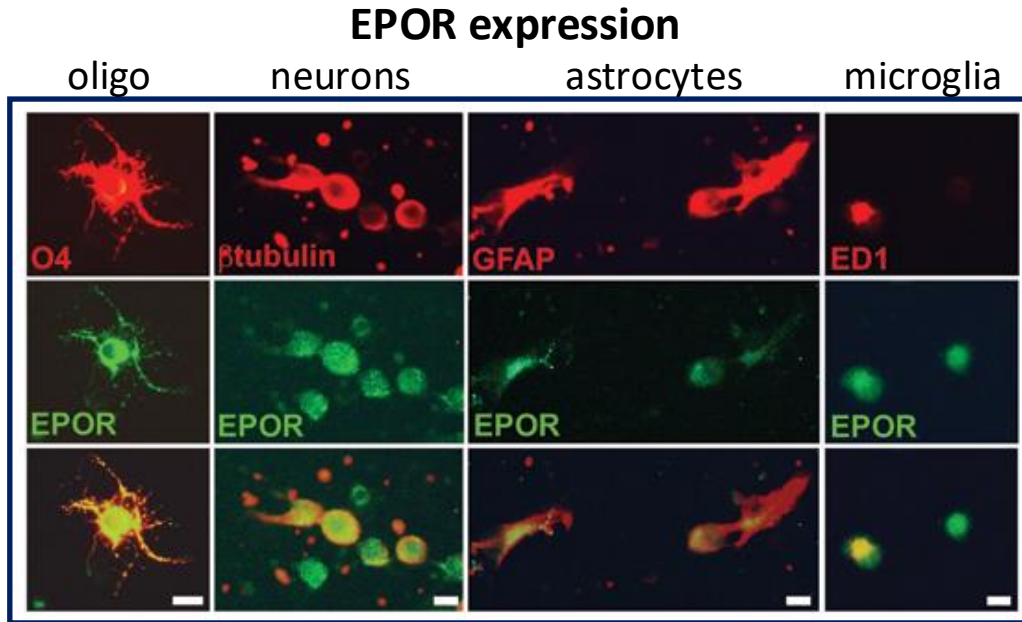
Erythropoietin in Neurodevelopment & Repair

Essential for CNS development and repair

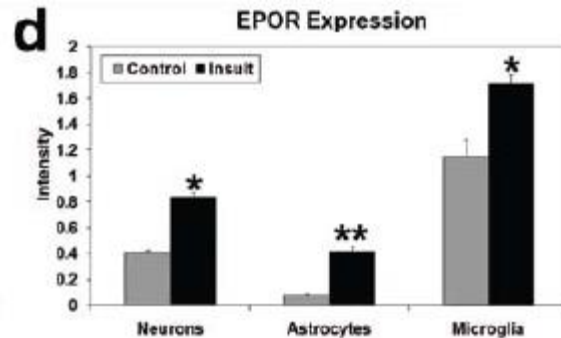
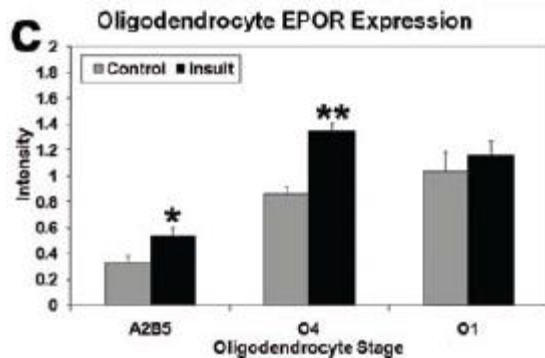
- Balance of ligand and receptors on most neural cells
- Present prenatally
- Marked postnatal reduction
- Required for repair processes



EPO Repairs Functional Outcomes in Perinatal Brain Injury



- Sustained deficits in adult rodents after early brain injury
 - Impaired gait
 - Seizure threshold
- Injury increases EPOR expression in neural cells



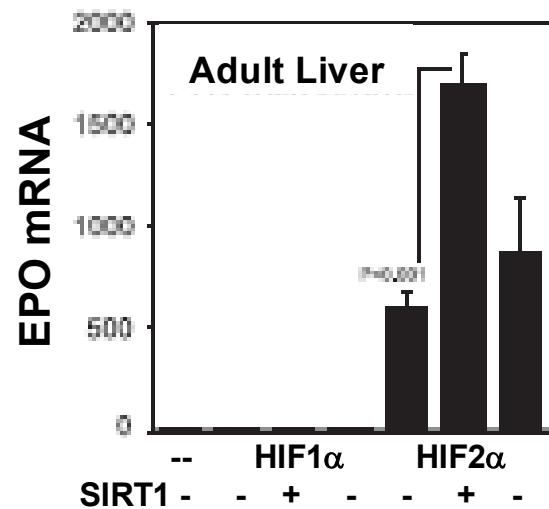
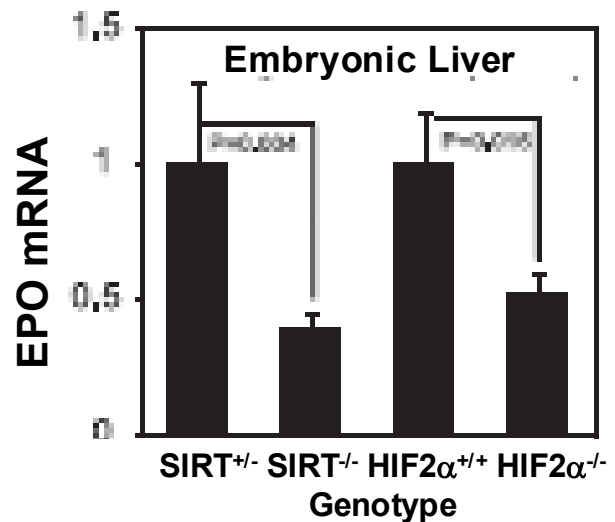
Mazur, ...Robinson, J Neurosurg Peds 2010
Jantzie, ...Robinson, Peds Res 2013
Jantzie, ... Robinson, Cereb Cort 2014
Jantzie, ... Robinson Mol Cell Neurosci, 2014
Jantzie, ...Robinson, Dev Neurosci 2015
Robinson et al, Exp Neurol 2018

Melatonin Enhances Sirt1

- EPO requires SIRT1 for protective effects of signaling

Regulation of Hypoxia-Inducible Factor 2 α Signaling by the Stress-Responsive Deacetylase Sirtuin 1

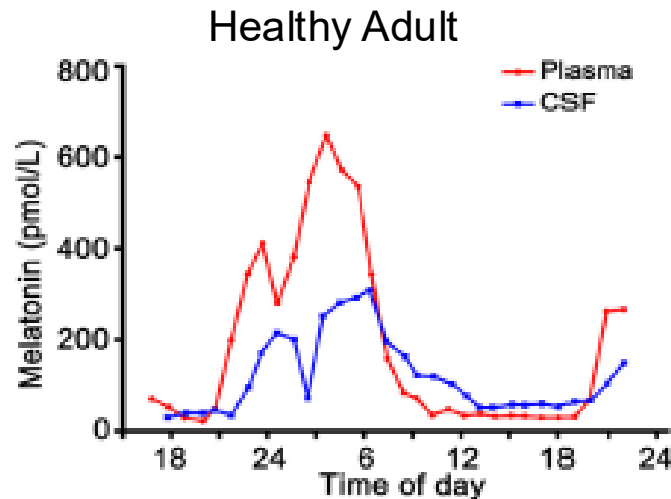
Elhadji M. Dioum,^{1,2*} Rui Chen,^{1,2*} Matthew S. Alexander,² Quiyang Zhang,² Richard T. Hogg,² Robert D. Gerard,^{2,3} Joseph A. Garcia^{1,2†}



Rationale for Melatonin

Why melatonin is :

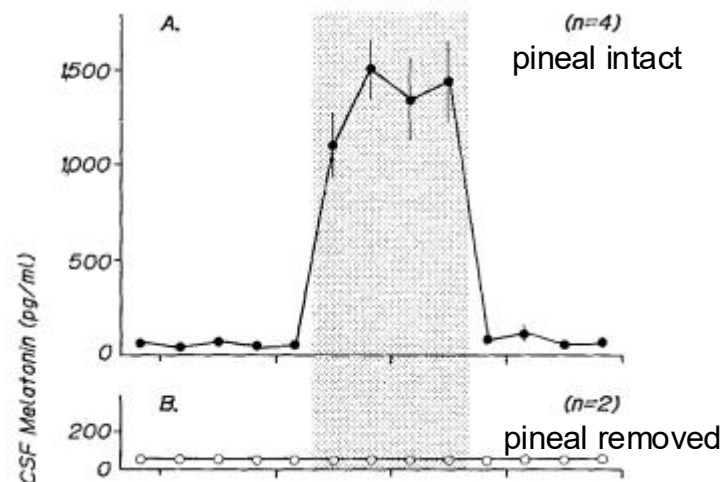
- High levels in ventricular CSF
- High levels at night when glymphatic system active



In newborns, replenishing deficit for mitochondrial receptors

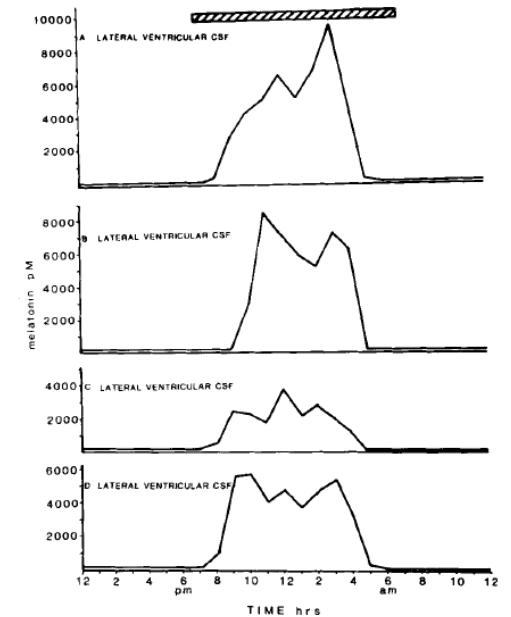
Presence of a Distinct 24-Hour Melatonin Rhythm in the Ventricular Cerebrospinal Fluid of the Goat

Nobue Kanematsu, Yuji Mori, Shinji Hayashi, and Kunio Hoshino



Evidence of High Concentrations of Melatonin in Lateral Ventricular Cerebrospinal Fluid of Sheep

Paul F. Shaw, David J. Kennaway, and Robert F. Seamark

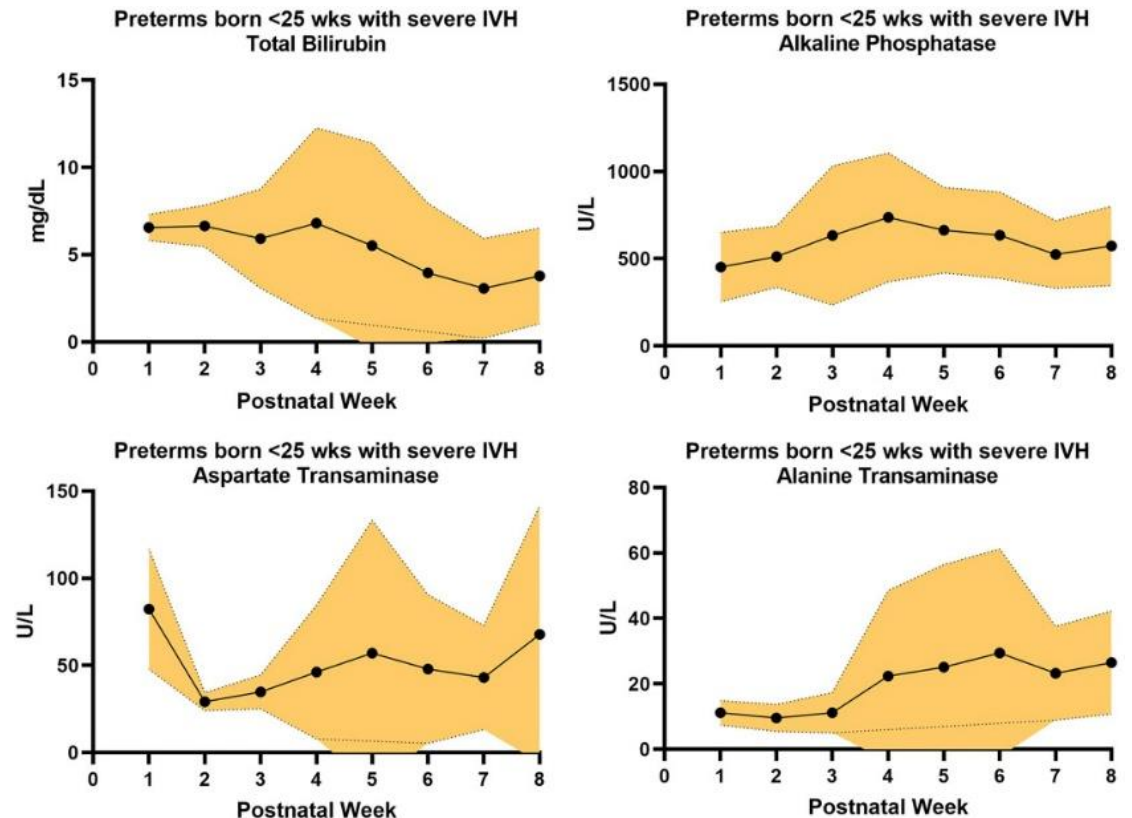


Hedlund et al, Science 1977
Shaw et al, J Pineal Res 1989
Kanematsu, J Pineal Res 1989
Bruce et al, J Clin Endo Metab 1991
Tricoire et al, Endocrin 2002
Longatti et al, J Pineal Res 2006
Reiter et al, J Pineal Res 2016
Reiter et al, Biochem Biophys Res Comm 2022

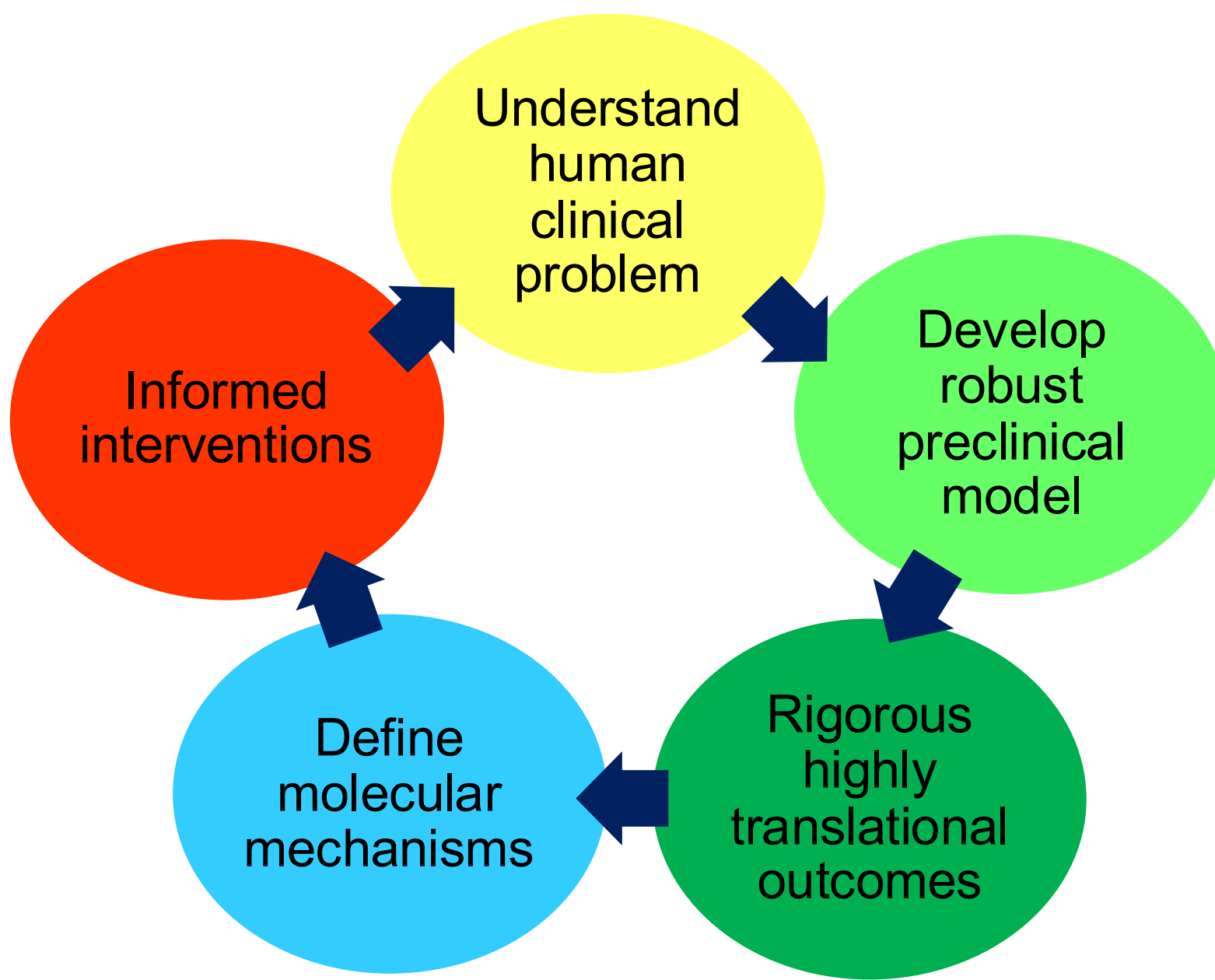
SCEMPI: Study Design

- Composite of 5 elements:
 1. Death due to all causes
 2. *Melatonin*: liver test nomograms
 3. *Erythropoietin*: hypertension
 4. *Erythropoietin*: thrombus
 5. *Erythropoietin*: polycythemia

Liver test nomograms



**What about my child
living with cerebral palsy?**



Pathway to Interventions

Rationale for HIF Stabilizer Vadadustat

